Lab Sheet 2

Tasks

1. Create a script.js file. In it, define a JavaScript array of product objects (id, name, price, image) representing your store's catalog — at least 8 products.
2. On products.html, use JavaScript to dynamically render the product cards from your array into the page (instead of hardcoded HTML blocks) using DOM methods like createElement / innerHTML with template literals.
3. Implement an 'Add to Cart' click handler on each product card that adds the product to a cart array (store it in a JS variable, and also persist it using localStorage so it survives a page refresh).
4. On cart.html, read the cart from localStorage and dynamically render each item with a working quantity input and a 'Remove' button.
5. Write a function calculateTotal() that recalculates and displays the cart's grand total whenever quantity changes or an item is removed.
6. On checkout.html, add form validation using JavaScript: Name and Address must not be empty, Pincode must be exactly 6 digits, Phone must be exactly 10 digits. Show inline error messages instead of using alert().
7. On successful validation, show a simple confirmation message on the page (e.g., 'Order placed!') and clear the cart from localStorage.
8. Add a small script on every page to update a 'cart item count' badge next to the Cart link in the navbar, read from localStorage.

Solutions-> index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Technical Store</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<nav>
    <a href="index.html">Home</a>
    <a href="products.html">Products</a>
    <a href="cart.html">Cart <span id="cart-count">0</span></a>
    <a href="checkout.html">Checkout</a>
</nav>
<header>
    <h1>Welcome to Technical Store</h1>
    <p>Your one-stop shop for quality tech products.</p>
```

```
    <a href="products.html">Shop Now</a>
</header>
<section>
    <h2>Featured Products</h2>
    <p>Explore our latest collection of technology products.</p>
    <a href="products.html">View All Products</a>
</section>
<script src="script.js"></script>
</body>
</html>
```

products.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Technical Store - Products</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<nav>
    <a href="index.html">Home</a>
    <a href="products.html">Products</a>
    <a href="cart.html">Cart <span id="cart-count">0</span></a>
    <a href="checkout.html">Checkout</a>
</nav>
<h1>Our Products</h1>
<div id="product-container"></div>
<script src="script.js"></script>
</body>
</html>
```

checkout.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Technical Store - Checkout</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
```

```
<nav>
    <a href="index.html">Home</a>
    <a href="products.html">Products</a>
    <a href="cart.html">Cart <span id="cart-count">0</span></a>
    <a href="checkout.html">Checkout</a>
</nav>
<h1>Checkout</h1>
<form id="checkout-form">
    <label>Name</label>
    <input type="text" id="name">
    <span id="name-error"></span>
    <label>Address</label>
    <textarea id="address"></textarea>
    <span id="address-error"></span>
    <label>Pincode</label>
    <input type="text" id="pincode" maxlength="6">
    <span id="pincode-error"></span>
    <label>Phone</label>
    <input type="text" id="phone" maxlength="10">
    <span id="phone-error"></span>
    <button type="submit">Place Order</button>
</form>
<h2 id="confirmation"></h2>
<script src="script.js"></script>
</body>
</html>
```

cart.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Technical Store - Cart</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<nav>
    <a href="index.html">Home</a>
    <a href="products.html">Products</a>
    <a href="cart.html">Cart <span id="cart-count">0</span></a>
    <a href="checkout.html">Checkout</a>
</nav>
```

```
<h1>Shopping Cart</h1>
<div id="cart-container"></div>
<h2>Grand Total: <span id="cart-total">₹0</span></h2>
<a href="checkout.html">Proceed to Checkout</a>
<script src="script.js"></script>
</body>
</html>
```

style.css

```
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}
body{
    font-family:Arial,sans-serif;
    background:#f5f7fa;
    color:#222;
}
nav{
    background:#1f2937;
    padding:18px 40px;
    display:flex;
    align-items:center;
    gap:30px;
}
nav a{
    color:white;
    text-decoration:none;
    font-size:17px;
}
nav a:hover{
    color:#38bdf8;
}
#cart-count{
    background:#ef4444;
    color:white;
    padding:3px 7px;
    border-radius:50%;
    font-size:12px;
    margin-left:3px;
}
header{
    text-align:center;
```

```
    padding:100px 20px;
    background:linear-gradient(135deg,#dbeafe,#eff6ff);
}
header h1{
    font-size:48px;
    margin-bottom:20px;
}
header p{
    font-size:20px;
    margin-bottom:30px;
    color:#555;
}
header a{
    display:inline-block;
    padding:13px 28px;
    background:#2563eb;
    color:white;
    text-decoration:none;
    border-radius:6px;
    font-size:17px;
}
header a:hover{
    background:#1d4ed8;
}
section{
    text-align:center;
    padding:60px 20px;
}
section h2{
    font-size:32px;
    margin-bottom:15px;
}
section p{
    color:#666;
    font-size:17px;
    margin-bottom:25px;
}
section a{
    display:inline-block;
    padding:10px 20px;
    background:#111827;
    color:white;
    text-decoration:none;
```

```
    border-radius:5px;
}
section a:hover{
    background:#374151;
}
#product-container{
    max-width:1200px;
    margin:40px auto;
    padding:0 20px;
    display:grid;
    grid-template-columns:repeat(4,1fr);
    gap:25px;
}
.product-card{
    background:white;
    border-radius:10px;
    padding:15px;
    box-shadow:0 3px 12px rgba(0,0,0,0.1);
    text-align:center;
    transition:transform 0.2s;
}
.product-card:hover{
    transform:translateY(-5px);
}
.product-card img{
    width:100%;
    height:200px;
    object-fit:cover;
    border-radius:8px;
}
.product-card h3{
    margin:15px 0 10px;
}
.product-card p{
    font-size:20px;
    font-weight:bold;
    margin-bottom:15px;
}
.product-card button{
    background:#2563eb;
    color:white;
    border:none;
    padding:10px 18px;
```

```
    border-radius:5px;
    cursor:pointer;
}
.product-card button:hover{
    background:#1d4ed8;
}
#cart-container{
    max-width:900px;
    margin:40px auto;
    padding:0 20px;
}
.cart-item{
    display:flex;
    align-items:center;
    gap:25px;
    background:white;
    padding:20px;
    margin-bottom:20px;
    border-radius:10px;
    box-shadow:0 3px 10px rgba(0,0,0,0.1);
}
.cart-item img{
    width:140px;
    height:120px;
    object-fit:cover;
    border-radius:8px;
}
.cart-item input{
    width:60px;
    padding:8px;
    margin:10px;
}
.cart-item button{
    background:#dc2626;
    color:white;
    border:none;
    padding:8px 15px;
    border-radius:5px;
    cursor:pointer;
}
.cart-item button:hover{
    background:#b91c1c;
}
```

```css
#cart-total{
    font-weight:bold;
    color:#2563eb;
}
form{
    max-width:500px;
    margin:40px auto;
    padding:30px;
    background:white;
    border-radius:10px;
    box-shadow:0 3px 15px rgba(0,0,0,0.1);
    display:flex;
    flex-direction:column;
}
form label{
    margin-top:10px;
    margin-bottom:5px;
    font-weight:bold;
}
form input,
form textarea{
    padding:10px;
    border:1px solid #ccc;
    border-radius:5px;
    font-size:15px;
}
form textarea{
    min-height:100px;
    resize:vertical;
}
form span{
    color:#dc2626;
    font-size:14px;
    margin-top:3px;
}
form button{
    margin-top:20px;
    padding:12px;
    background:#2563eb;
    color:white;
    border:none;
    border-radius:5px;
    cursor:pointer;
```

```
    font-size:16px;
}
form button:hover{
    background:#1d4ed8;
}
#confirmation{
    text-align:center;
    color:green;
    margin:20px;
}
@media(max-width:900px){
    #product-container{
        grid-template-columns:repeat(2,1fr);
    }
}
@media(max-width:600px){
    nav{
        padding:15px;
        gap:15px;
        flex-wrap:wrap;
    }
    #product-container{
        grid-template-columns:1fr;
    }
    .cart-item{
        flex-direction:column;
        text-align:center;
    }
    header h1{
        font-size:35px;
    }
}
```

script.js

```
const products=[
{id:1,name:"Wireless
Headphones",price:1499,image:"https://images.unsplash.com/photo-1505740
420928-5e560c06d30e"},
{id:2,name:"Smart
Watch",price:2499,image:"https://images.unsplash.com/photo-152327533568
4-37898b6baf30"},
{id:3,name:"Running
Shoes",price:1999,image:"https://images.unsplash.com/photo-1542291026-7
eec264c27ff"},
```

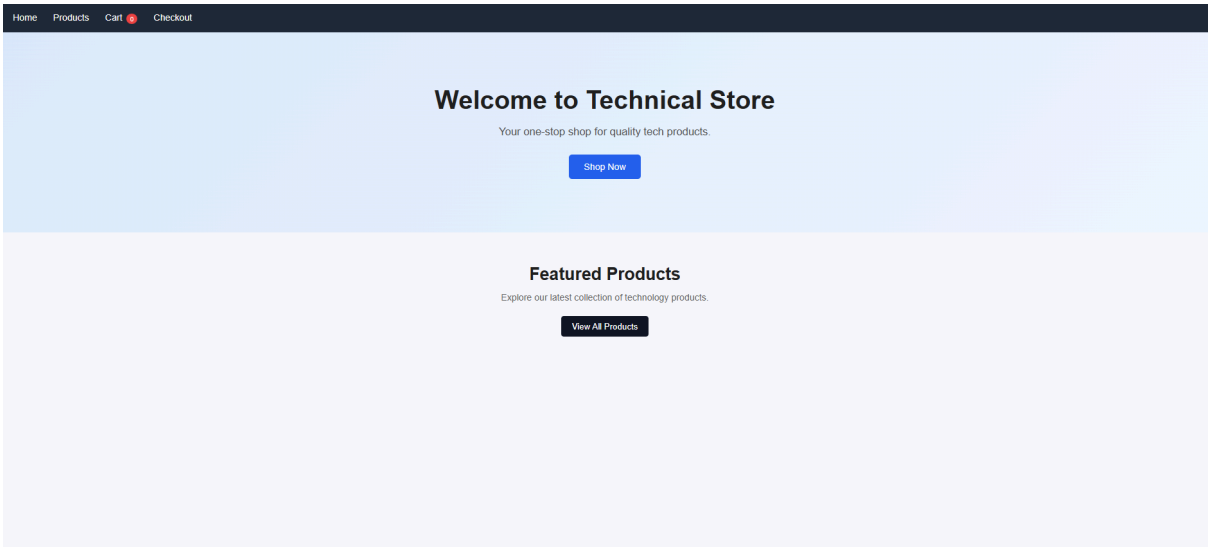
```
{id:4,name:"Laptop
Backpack",price:1299,image:"https://images.unsplash.com/photo-155306240
7-98eeb64c6a62"},
{id:5,name:"Bluetooth
Speaker",price:1799,image:"https://images.unsplash.com/photo-1608043152
269-423dbba4e7e1"},
{id:6,name:"Gaming
Mouse",price:999,image:"https://images.unsplash.com/photo-1527814050087
-3793815479db"},
{id:7,name:"Mechanical
Keyboard",price:2299,image:"https://images.unsplash.com/photo-158782974
1301-dc798b83add3"},
{id:8,name:"Sunglasses",price:799,image:"https://images.unsplash.com/ph
oto-1511499767150-a48a237f0083"}
];
let cart=JSON.parse(localStorage.getItem("cart"))||[];
function updateCartCount(){
    const count=document.getElementById("cart-count");
    if(count){
        const
totalItems=cart.reduce((total,item)=>total+item.quantity,0);
        count.textContent=totalItems;
    }
}
function saveCart(){
    localStorage.setItem("cart",JSON.stringify(cart));
    updateCartCount();
}
function addToCart(id){
    const product=products.find(product=>product.id===id);
    const existingItem=cart.find(item=>item.id===id);
    if(existingItem){
        existingItem.quantity++;
    }else{
        cart.push({
            id:product.id,
            name:product.name,
            price:product.price,
            image:product.image,
            quantity:1
        });
    }
    saveCart();
```

```
}
function renderProducts(){
    const container=document.getElementById("product-container");
    if(!container)return;
    container.innerHTML="";
    products.forEach(product=>{
        const card=document.createElement("div");
        card.className="product-card";
        card.innerHTML=`
            <img src="${product.image}" alt="${product.name}">
            <h3>${product.name}</h3>
            <p>₹${product.price}</p>
            <button onclick="addToCart(${product.id})">Add to
Cart</button>
        `;
        container.appendChild(card);
    });
}
function renderCart(){
    const container=document.getElementById("cart-container");
    const totalElement=document.getElementById("cart-total");
    if(!container)return;
    container.innerHTML="";
    if(cart.length===0){
        container.innerHTML="<p>Your cart is empty.</p>";
        if(totalElement){
            totalElement.textContent="₹0";
        }
        return;
    }
    cart.forEach(item=>{
        const cartItem=document.createElement("div");
        cartItem.className="cart-item";
        cartItem.innerHTML=`
            <img src="${item.image}" alt="${item.name}">
            <div>
                <h3>${item.name}</h3>
                <p>Price: ₹${item.price}</p>
                <label>Quantity:</label>
                <input type="number" min="1" value="${item.quantity}"
onchange="updateQuantity(${item.id},this.value)">
                <button
onclick="removeFromCart(${item.id})">Remove</button>
```

```
            </div>
        `;
        container.appendChild(cartItem);
    });
    calculateTotal();
}
function updateQuantity(id,quantity){
    const item=cart.find(item=>item.id===id);
    if(item){
        item.quantity=Math.max(1,parseInt(quantity)||1);
        saveCart();
        renderCart();
    }
}
function removeFromCart(id){
    cart=cart.filter(item=>item.id!==id);
    saveCart();
    renderCart();
}
function calculateTotal(){
    const totalElement=document.getElementById("cart-total");
    if(!totalElement)return;
    const total=cart.reduce((sum,item)=>{
        return sum+(item.price*item.quantity);
    },0);
    totalElement.textContent=`₹${total}`;
}
function validateCheckout(event){
    event.preventDefault();
    const name=document.getElementById("name");
    const address=document.getElementById("address");
    const pincode=document.getElementById("pincode");
    const phone=document.getElementById("phone");
    const nameError=document.getElementById("name-error");
    const addressError=document.getElementById("address-error");
    const pincodeError=document.getElementById("pincode-error");
    const phoneError=document.getElementById("phone-error");
    const confirmation=document.getElementById("confirmation");
    nameError.textContent="";
    addressError.textContent="";
    pincodeError.textContent="";
    phoneError.textContent="";
    confirmation.textContent="";
```

```
    let valid=true;
    if(name.value.trim()===""){
        nameError.textContent="Name is required.";
        valid=false;
    }
    if(address.value.trim()===""){
        addressError.textContent="Address is required.";
        valid=false;
    }
    if(!/^\d{6}$/.test(pincode.value)){
        pincodeError.textContent="Pincode must be exactly 6 digits.";
        valid=false;
    }
    if(!/^\d{10}$/.test(phone.value)){
        phoneError.textContent="Phone must be exactly 10 digits.";
        valid=false;
    }
    if(valid){
        confirmation.textContent="Order placed!";
        cart=[];
        localStorage.removeItem("cart");
        updateCartCount();
        document.getElementById("checkout-form").reset();
    }
}
document.addEventListener("DOMContentLoaded",()=>{
    updateCartCount();
    renderProducts();
    renderCart();
    const checkoutForm=document.getElementById("checkout-form");
    if(checkoutForm){
        checkoutForm.addEventListener("submit",validateCheckout);
    }
});
```

Output ->


Home
Products
Cart 0
Checkout
Welcome to Technical Store
Your one-stop shop for quality tech products.
Shop Now
Featured Products
Explore our latest collection of technology products.
View All Products